Fast Kriging Uncertainty Quantification and LSTM: Tools for the Spatial and Temporal Aspects of Hydrologic Modeling

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Outline

Motivation

Figure: Water in Lost Man canal passes underneath Highway 82 on Independence Pass. Courtesy of Aspen Public Radio.

Figure: Grand Lake (formerly Spirit Lake) in Colorado. Courtesy of Mountain Town Magazine.

Hydrologic Modeling

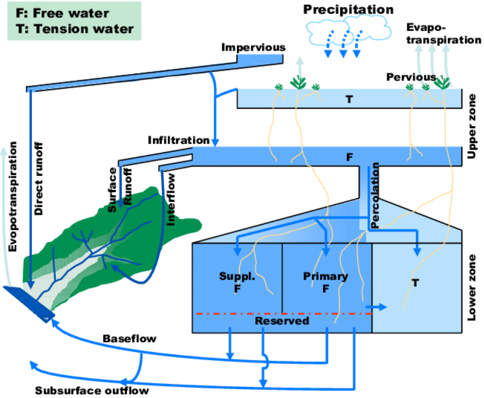

Figure: The Sacramento Soil Moisture Accounting (SAC-SMA) Scheme [1]. This model does not have a snow module.

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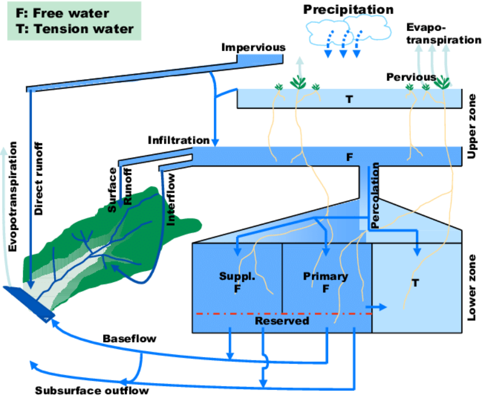

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Basin/Catchment: portion of land drained by a river and its tributaries.

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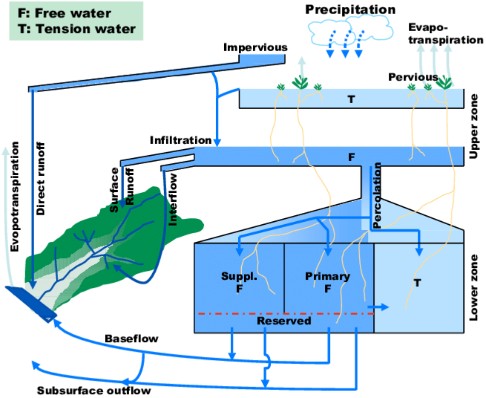

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Spatial Aspects (w/ Bailey et al.):

- Precipitation field
- Flood plain
- Total snow water equivalent (SWE) or snowcover

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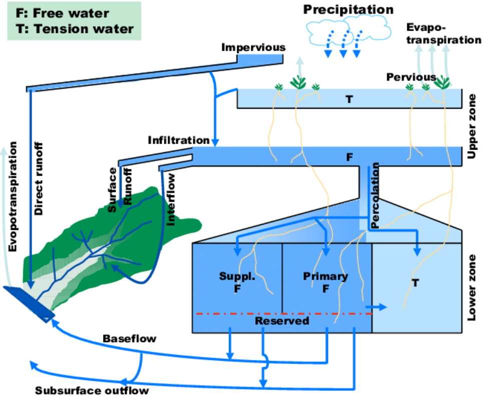

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Time Aspects (w/ Kratzert et al.):

- Direct runoff
- Groundwater storage
- Rainfall duration

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1. Background
2. Fast Kriging Uncertainty Approx.
3. LSTM for Rainfall-Runoff Modeling
4. Conclusion

Technical Background: Spatial Model

"Adapting conditional simulation using circulant embedding for irregularly spaced spatial data" [2]

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Observational Model:

$$y_i = \mu(\mathbf{s}_i) + g(\mathbf{s}_i) + \epsilon_i \tag{1}$$

$y_i =$ the i-th observation, $i = 1, \ldots, n \in \mathbb{N}$.

$\mu(\mathbf{s}_i) =$ the mean (aka. *fixed effect*, *parametric* part, or *global* part) at the i-th location $\mathbf{s}_i$.

$g(\mathbf{s}_i) =$ the mean zero, stochastic *spatial process*(aka. *nonparametric* part, or *local* part) at $\mathbf{s}_i$.

$\epsilon_i =$ white noise term when observing $g(\mathbf{s}_i)$. $\epsilon_i \overset{\text{iid}}{\sim} N(0, \tau^2)$, τ^2 nugget variance.

Technical Background: Universal Kriging & Uncertainty

Simple Kriging
For where μ is known or $\mu = 0$:

$$\begin{aligned} \hat{g}(\mathbf{s}^*) &= \mathbf{k}^{\mathrm{T}} \left(\mathbf{K} + \tau^2 I\right)^{-1} \mathbf{y} \\ \mathrm{Var}\left(\hat{g}(\mathbf{s}^*) - g(\mathbf{s}^*)\right) &= \mathrm{Var}(g(\mathbf{s}^*)) - \mathbf{k}^{\mathrm{T}} \left(\mathbf{K} + \tau^2 I\right)^{-1} \mathbf{k} \end{aligned} \tag{2}$$

$\mathbf{k}_i = k(\mathbf{s}^*, \mathbf{s}_i^O)$ for a covariance function k, and $\{\mathbf{K}\}_{ij} = k(\mathbf{s}_i^O, \mathbf{s}_j^O)$.

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Universal Kriging

For estimating $f = \mu + g$ at new location $\mathbf{s}^*$:

$$\hat{f}(\mathbf{s}^*) = \mathbf{x}(\mathbf{s}^*)^{\mathrm{T}}\hat{\boldsymbol{\beta}} + \mathbf{k}^{\mathrm{T}}\mathrm{M}^{-1}\left(\mathbf{y}-\mathbf{X}\hat{\boldsymbol{\beta}}\right)$$

Where $\mathrm{M} = \mathbf{K} + \tau^2 I = \mathrm{Cov}(\mathbf{y}, \mathbf{y})$.

Technical Background: Circulant Embedding (CE)

Goal: Simulate the mean zero Gaussian process at locations $\mathbf{s}$, $g(\mathbf{s}) \sim \mathrm{MVN}(0, \Sigma)$.
With $\mathbf{s} = [\mathbf{s}_1, \ldots, \mathbf{s}_M]$, $\{\Sigma\}_{ij} = k(\mathbf{s}_i, \mathbf{s}_j)$ and k a covariance function.

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- Exact
- Do not assume:
  - stationarity (translation invariance)
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- $\mathcal{O}(n^3)$

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Embed Σ into $\mathsf{M} = \begin{bmatrix} \Sigma & \mathsf{M}_{12} \\ \mathsf{M}_{21} & \mathsf{M}_{22} \end{bmatrix}$, $\mathsf{M} = UDU^H$
Form $D^{1/2}$ w/ eigenvals $\mathbf{d} = U^H \mathsf{M}_{1,\cdot} = \mathcal{F}(\mathsf{M}_{1,\cdot})$
Get $g(\mathbf{s})$ from first M entries of

$$UD^{1/2}U^H\boldsymbol{\varepsilon} = \mathcal{F}^{-1}(D^{1/2}\mathcal{F}(\boldsymbol{\varepsilon}))$$

Bailey et al. 2022: Original Conditional Simulation

Data: $\mathbf{y}$ at n spatial locations $\mathcal{S}^O = \{\mathbf{s}_1^O, \mathbf{s}_2^O, \ldots, \mathbf{s}_n^O\}$.
Goal: Quantify uncertainty of predictions $\hat{g}$ on evenly spaced grid $\mathcal{S}^G = \{\mathbf{s}_1^G, \mathbf{s}_2^G, \ldots, \mathbf{s}_M^G\}$.

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Input: Spatial data $\mathbf{y}$, their locations $\mathcal{S}^O$, and prediction grid locations $\mathcal{S}^G$.
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For $j = 1 : l$

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- Form synthetic data $\mathrm{y}_i^S = g^S(\mathbf{s}_i^O) + \epsilon_i^S$, for $\mathbf{s}_i^O \in \mathcal{S}^O$, for $\epsilon_i^S \stackrel{\text{iid}}{\sim} N(0, \tau^2)$ and $i = 1, \ldots, n$.

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Bailey et al. 2022: Neighborhood Order $n_p = 2$ Example

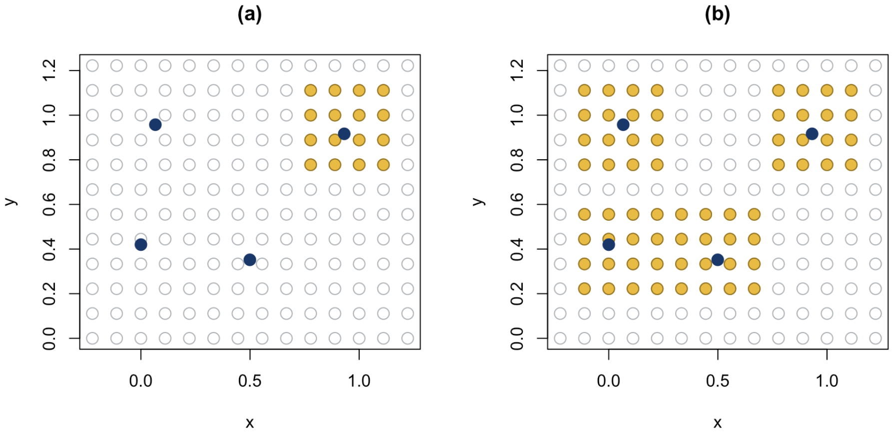

Figure: Example of 4 observations $\mathbf{y} = [y_1, y_2, y_3, y_4]^\mathrm{T}$ labeled as dark blue dots at $\mathcal{S}^O$. $\mathcal{S}^G$ are shown as gray circles. (a) shows $\mathcal{S}^G_{\eta(1)}$ and (b) shows $\mathcal{S}^G_{\boldsymbol{\eta}}$ in yellow .

Bailey et al. 2022: Local Kriging

Choose the n_p order neighborhood of points near the i-th observation $\mathbf{s}_i^O$ on the grid as $\mathcal{S}_{\eta(i)}^G \subset \mathcal{S}^G$ to approximate off-grid simulated values.

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Input: Spatial data $\mathbf{y}$, their locations $\mathcal{S}^O$, prediction grid locations $\mathcal{S}^G$, neighborhood order n_p.

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2. $g^S(\mathbf{s}_i)$ and $g^S(\mathbf{s}_j)$ are independent conditional on $g^S(\mathcal{S}^G_{\eta(i)}) \cup g^S(\mathcal{S}^G_{\eta(j)})$. Reasonable due to fine resolution of CE.
    - Allows for parallel simulation of the process at observation locations
    - Correlation falls to $< 5\%$ when 20+ units apart

Bailey et al. 2022: Approximate Grid Embedding

Create surrogate observations by shifting information from $\mathbf{s}_i^O$ to its nearest neighbors $\mathcal{S}_{\eta(i)}^G$ on grid.

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Bailey et al. 2022: Approximate Grid Embedding

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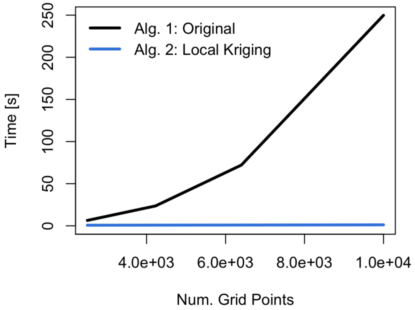
- Obtain the j-th conditional simulation as $\mathbf{v}_j = \hat{g}(\mathcal{S}^G) + \left(g^S(\mathcal{S}^G) - \hat{g}^S(\mathcal{S}^G)\right)$.

End

Bailey et al. 2022: Conclusion

Results



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Bailey et al. 2022: Conclusion

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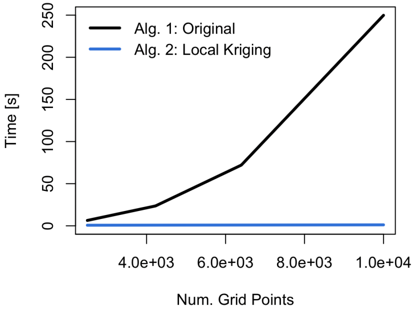

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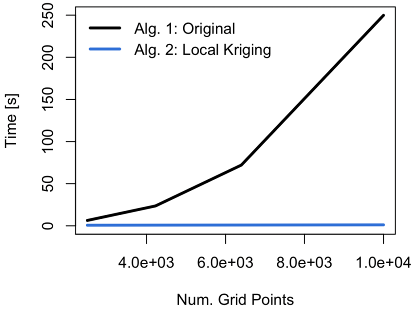

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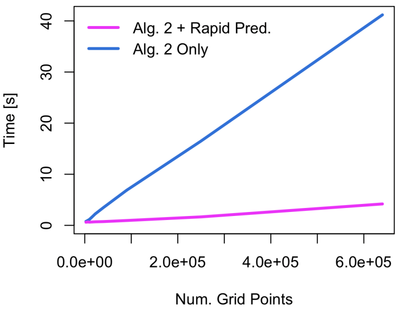
Current/Future Work

Rapid Approximate Prediction Method



- Quantify uncertainty for Rapid Pred.
- Fix problems with CE ill-conditioning

Outline

Technical Background: Time Series

Goal: Suppose we have runoff y_t, and some forcings $\mathbf{x}_t$ for $t = 1, \ldots, n$. Predict y_{n+1}.

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Auto-regressive moving average models, ARMA(p, q)

$$y_{n+1} = \sum_{i=1}^{p} \Phi_i y_{n+1-i} + \sum_{j=1}^{q} \gamma_i w_{n+1-j} + w_{n+1}$$

$$\begin{aligned} \Phi_i, \gamma_j &= \text{ weights for AR}(p) \text{ \& MA}(q). \\ w_{n+1} &= \text{ white noise.} \end{aligned}$$

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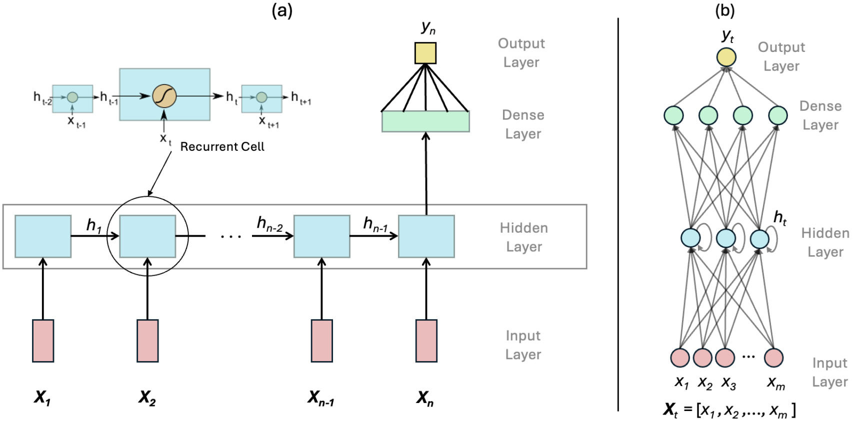
Problems:

- No long term memory
- Difficulties w/ nonlinear features

Technical Background: Recurrent Neural Net (RNN)



Deals with non-linear features better:

$$\mathbf{h}_t = g(\mathbf{W}\mathbf{x}_t + \mathbf{U}\mathbf{h}_{t-1})$$

Figure: (a) Example one-layer recurrent neural network. (b) Cross-section for a single time of inputs $\mathbf{x}_t$. 3 hidden units per layer.

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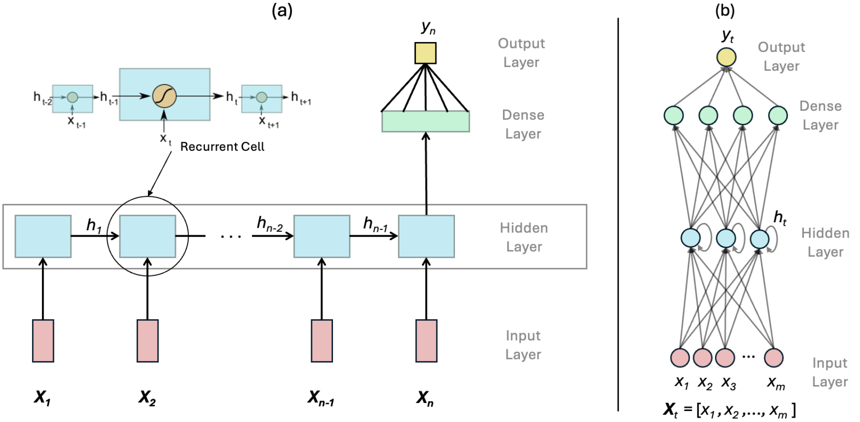

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Problem:
Effectively no long-term memory – hidden state memory decays exponentially

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Hochreiter & Schmidhuber [4] proposed idea in 1997: gating/controlling long-term information flow.

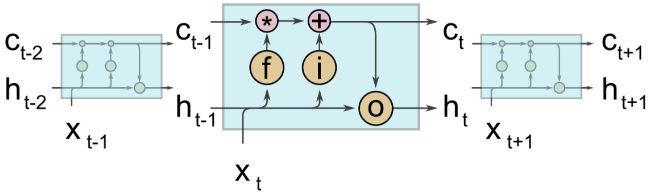

Figure: LSTM unit at time step t

Has memory cell state $\mathbf{c}_t$ for long-term information with 3 gates.

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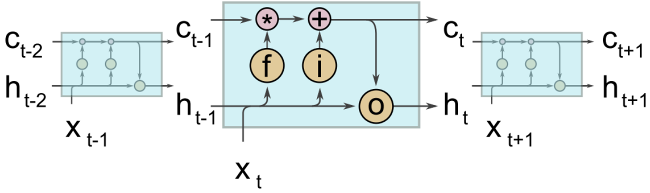

Figure: LSTM unit at time step t

$$\begin{aligned}
\textbf{forget gate:}\quad & f_t = \sigma(\mathbf{W}_f \mathbf{x}_t + \mathbf{U}_f \mathbf{h}_{t-1}) \\
\text{update vector:}\quad & \tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_{\tilde{\mathbf{c}}} \mathbf{x}_t + \mathbf{U}_{\tilde{\mathbf{c}}} \mathbf{h}_{t-1}) \\
\textbf{input gate:}\quad & i_t = \sigma(\mathbf{W}_i \mathbf{x}_t + \mathbf{U}_i \mathbf{h}_{t-1}) \\
\text{cell state:}\quad & \mathbf{c}_t = f_t \odot \mathbf{c}_{t-1} + i_t \odot \tilde{\mathbf{c}}_t \\
\textbf{output gate:}\quad & o_t = \sigma(\mathbf{W}_o \mathbf{x}_t + \mathbf{U}_o \mathbf{h}_{t-1}) \\
\text{hidden state:}\quad & \mathbf{h}_t = \tanh(\mathbf{c}_t) \odot o_t
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Kratzert et al. 2018: Background

"LSTM for Rainfall-Runoff Modeling" [5].

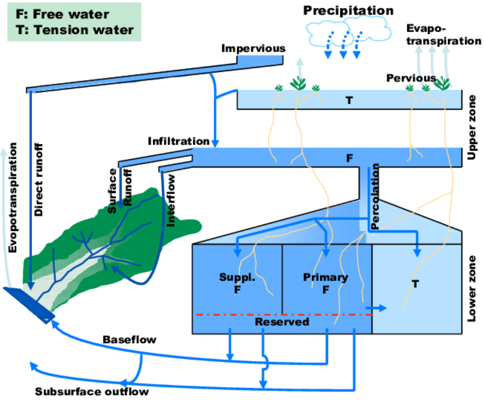

Figure: SAC-SMA Scheme [1]. Technically a conceptual model – simpler version of PBMs.

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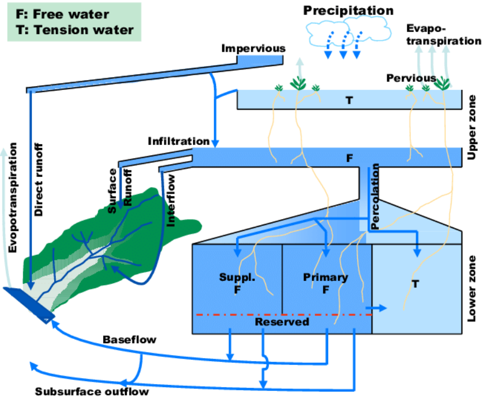

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Process-Based Models (PBMs):

- Based on physics or empirical relationships.
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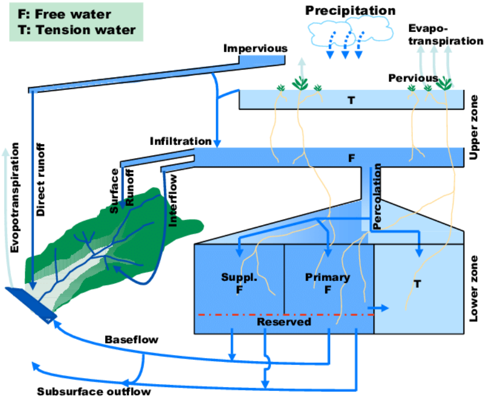

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Problems:

- Too big to run operationally.
- Lack of scale-relevant science.
- Poor prediction in "ungauged" catchments, etc.

Kratzert et al. 2018: Experimental Design

Goal: Examine LSTM's ability to model rainfall-runoff, w/ focus in arid regions and catchment-specific vs. regional pattern capabilities.

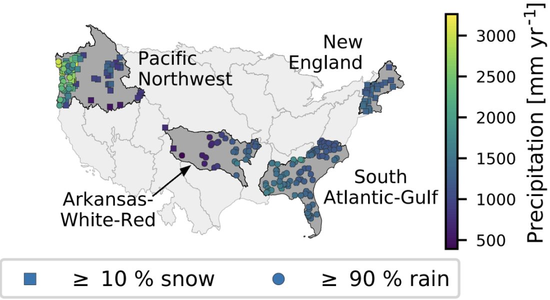

Figure: Mean annual precipitation of regions chosen for experiments [5]. There are 241 catchments total.

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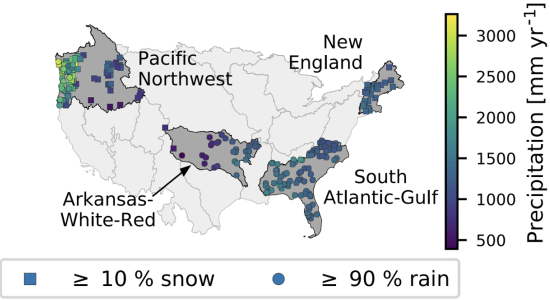

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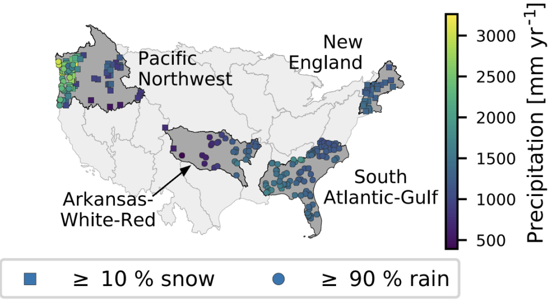

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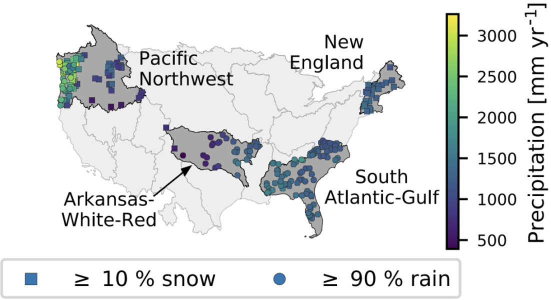

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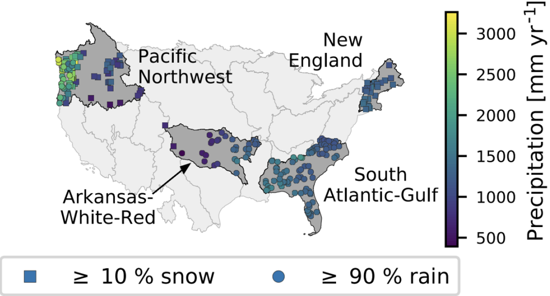

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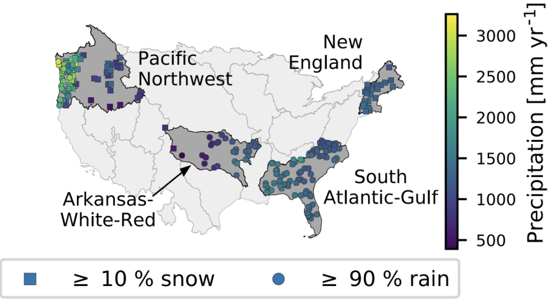

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Evaluation via Nash–Sutcliffe model efficiency coefficient (NSE):

$$NSE = 1 - \frac{\sum_t^T (y_t^{obs} - y_t^{sim})^2}{\sum_t^T (y_t^{obs} - \bar{y}^{obs})^2}$$

Kratzert et al. 2018: Architecture

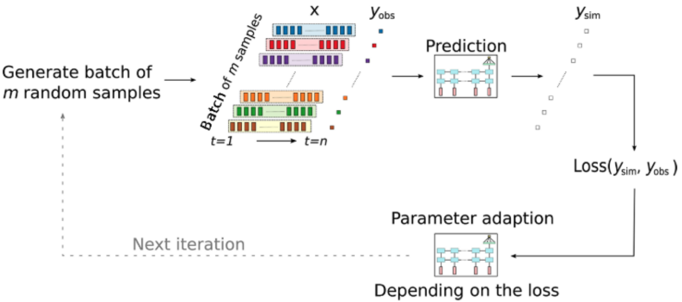

Figure: Each iteration of training [5].

- Drop out rate: 10%.
- Loss function: mean square error.
- Data: “Catchment Attributes for Large-Sample Studies” (CAMELS).

Kratzert et al. 2018: Results

Experiment 1:

- LSTM outperforms PBM benchmark slightly in dry basins and snow-influenced basins.
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NEURAL HYDROLOGY
Using Neural Networks in Hydrology

Extensions

Shen et al. proposed an alternative: differentiable process based models (dPBMs) [6].

Process Based Modeling:

$$y_{sim} = g^{\theta}(u, x, A)$$

where, calibrate $\theta = argmin(L(y_{sim}, y_{obs}))$

Purely Data-Driving ML Modeling:

$$y_{sim} = f^{W}(u, x, A)$$

where, optimize $W = argmin(L(y_{sim}, y_{obs}))$

Differentiable Process Based Modeling Approach:

$$y_{sim} = g^{\theta}(u, x, A)$$

where, $\theta = f^{W}(u, x, A)$ and optimize $W = argmin(L(y_{sim}, y_{obs}))$

Figure: u = the internal states, x = forcings, A = static or semi-static basin attributes. θ = small set of interpretable parameters. W = neural net weights. L = loss function.

Current/Future Work:

NOAA Coop. Institute for Research to Operations in Hydrology (CIROH): "Deep Learning models and data assimilation for the NextGen Global Prediction Sys. Framework" led by Dr. Andy Wood (Mines & NCAR) and Dr. Jonathan Frame (U. Alabama).

Desired Outcome: predict static θ_{CFE} to be used in CFE that is part of NextGen to replace WRF-Hydro.

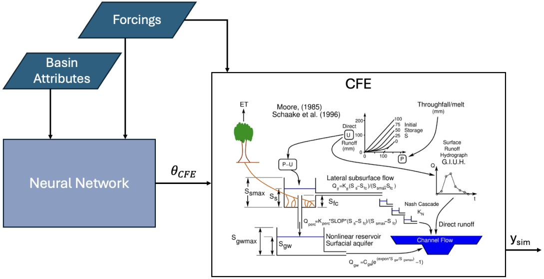

Figure: Scheme for differential version of the Conceptual Equivalent Model (CFE), functionally equivalent to the National Water Model (NWM).

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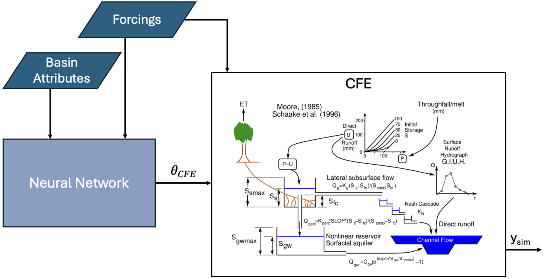

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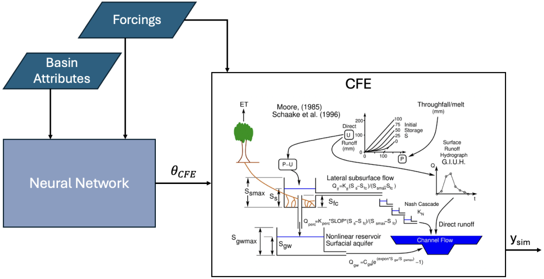

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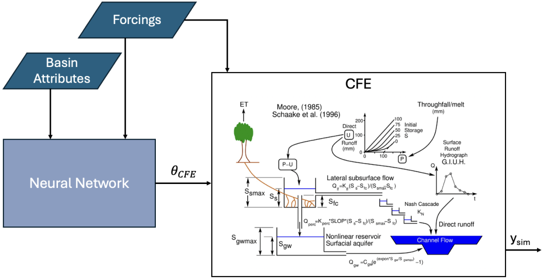

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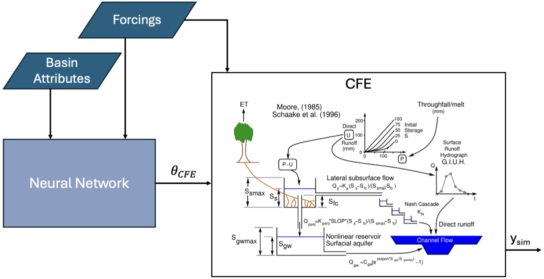

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- Better data products
- Data assimilation w/ dPBM
- Simple time-series model for θ or better ML architecture

Outline

Summary

Computational problems in statistics & ML
can hinder the advancement of hydrological sciences.

Professional Development

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- MATH 531: Theory of LM (Required, Sp25)
- MATH 540: Parallel Scientific Computing
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- AGU (Dec. 2025; New Orleans, LA)
- JSM2025, ML4Sci2025(?), Banff2026

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- ENVR 2024, Extremes 2024 (Poster) ✓
- Spatial Statistics (July 2025; Netherlands)
- AGU (Dec. 2025; New Orleans, LA)
- JSM2025, ML4Sci2025(?), Banff2026

Current Work

- Sustainability in labs (non-thesis) – expected Win24, Royal Society of Chemistry
- Rapid Approx. Prediction – expected Sp25, Computational Statistics
- dCFE/dPBM – hopefully Su25

Professional Development

Coursework

- MATH 535: Math Stat 2 (Required, Sp25)
- MATH 531: Theory of LM (Required, Sp25)
- MATH 540: Parallel Scientific Computing
- MATH 551: Computational Linear Algebra
- APPM 5510: Data Assimilation (Fa25)

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Summer Ideas

- Intern at Google Flood Hub
- Intern at NCAR or NOAA
- Intern at Argonne or Lawrence Berkeley
- Summer HPC courses/workshops

Thank you!

References

[1] Elizabeth Basha. "In-Situ Prediction on Sensor Networks Using Distributed Multiple Linear Regression Models". In: (Dec. 2010).

[2] Maggie D. Bailey, Soutir Bandyopadhyay, and Douglas Nychka. "Adapting conditional simulation using circulant embedding for irregularly spaced spatial data". In: *Stat* 11 (1 Dec. 2022). ISSN: 20491573. DOI: 10.1002/sta4.446.

[3] Doug Nychka, Soutir Bandyopadhyay, and William Kleiber. "Spatial Data Science (unpublished)". May 2023.

[4] Sepp Hochreiter and Jürgen Schmidhuber. "Long Short-Term Memory". In: *Neural Computation* 9 (8 Nov. 1997), pp. 1735–1780. ISSN: 08997667. DOI: 10.1162/neco.1997.9.8.1735.

[5] Frederik Kratzert et al. "Rainfall-runoff modelling using Long Short-Term Memory (LSTM) networks". In: *Hydrology and Earth System Sciences* 22 (11 Nov. 2018), pp. 6005–6022. ISSN: 16077938. DOI: 10.5194/hess-22-6005-2018.

[6] Chaopeng Shen et al. "Differentiable modelling to unify machine learning and physical models for geosciences". In: *Nature Reviews Earth and Environment* 4 (8 Aug. 2023), pp. 552–567. ISSN: 2662138X. DOI: 10.1038/s43017-023-00450-9.

Extra Technical Background: Simulation Methods

Simulation via Cholesky Decompositon
Cholesky decomposition: $\Sigma = BB^{\mathrm{T}}$
Obtain simulation: $g(\mathbf{s}) = B\boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathrm{MVN}(0, I)$

$$\mathbb{E}[B\boldsymbol{\epsilon}] = B\mathbb{E}[\boldsymbol{\epsilon}] = 0$$

$$\mathrm{Var}(B\boldsymbol{\epsilon}) = B\mathrm{Var}(\boldsymbol{\epsilon})B^{\mathrm{T}} = BIB^{\mathrm{T}} = \Sigma$$

Extra Technical Background: Circulant & DFT Matrices

Define $\underline{c} = [c_1, c_2, \ldots, c_N]$, $N \in \mathbb{N}$ and $c_1, c_2, \ldots, c_N \in \mathbb{R}$.

$$\mathsf{M} = \begin{bmatrix} c_1 & c_2 & c_3 & \cdots & c_N \\ c_N & c_1 & c_2 & \cdots & c_{N-1} \\ c_{N-1} & c_N & c_1 & \cdots & c_{N-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ c_2 & c_3 & c_4 & \cdots & c_1 \end{bmatrix}, \text{and } \mathsf{U} = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega & \omega^2 & \cdots & \omega^{N-1} \\ 1 & \omega^2 & \omega^4 & \cdots & \omega^{2(N-1)} \\ 1 & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{N-1} & \omega^{2(N-1)} & \cdots & \omega^{(N-1)^2} \end{bmatrix} \tag{3}$$

$\mathsf{M} =$ circulant matrix generated by $\underline{c}$.

$\mathsf{U} =$ unitary discrete Fourier transform (DFT) matrix with $\omega = e^{-2\pi i/N}$.

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Special property: $\mathsf{U}^H \underline{c} = \mathcal{F}(\underline{c})$ and $\mathsf{U}\underline{c} = \mathcal{F}^{-1}(\underline{c})$ [3].

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Eigendecompostion:

$$\mathsf{M} = \mathsf{U}D\mathsf{U}^H, \quad \mathsf{M}^{1/2} = \mathsf{U}D^{1/2}\mathsf{U}^H$$

Extra Technical Background: 1D Spatial Process Example

Goal: simulate g on equally spaced grids $\mathcal{S}^G = \{\mathbf{s}_1^G, \mathbf{s}_2^G, \ldots, \mathbf{s}_M^G\}$ w/ stationary covariance kernel k.

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Stationarity $\rightarrow \mathrm{Cov}(g(\mathbf{s}_i^G), g(\mathbf{s}_j^G)) = k(\mathbf{s}_i^G, \mathbf{s}_j^G) = \tilde{k}(\mathbf{s}_i^G - \mathbf{s}_j^G)$ for $i, j \in \{1, \ldots M\}$.

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$$\Sigma = \begin{bmatrix} \tilde{k}(0) & \tilde{k}(\Delta s) & \tilde{k}(2\Delta s) & \tilde{k}(3\Delta s) & \cdots & \tilde{k}((M-1)\Delta s) \\ \tilde{k}(\Delta s) & \tilde{k}(0) & \tilde{k}(\Delta s) & \tilde{k}(2\Delta s) & \cdots & \tilde{k}((M-2)\Delta s) \\ \tilde{k}(2\Delta s) & \tilde{k}(\Delta s) & \tilde{k}(0) & \tilde{k}(\Delta s) & \cdots & \tilde{k}((M-3)\Delta s) \\ \tilde{k}(3\Delta s) & \tilde{k}(2\Delta s) & \tilde{k}(\Delta s) & \tilde{k}(0) & \cdots & \tilde{k}((M-4)\Delta s) \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \tilde{k}((M-1)\Delta s) & \tilde{k}((M-2)\Delta s) & \tilde{k}((M-3)\Delta s) & \tilde{k}((M-4)\Delta s) & \ldots & \tilde{k}(0) \end{bmatrix}$$

Extra Technical Background: 1D Example Cont.

Suppose $\underline{c} = \begin{bmatrix} \tilde{k}(0) & \tilde{k}(\delta s) & \cdots & \tilde{k}((M-1)\delta s) & \tilde{k}((M-2)\delta s) & \cdots & \tilde{k}(2\delta s) & \tilde{k}(\delta s) \end{bmatrix} \in \mathbb{R}^{2M-2}$.
Then, *Embed* Σ into a circulant matrix:

$$\mathsf{M} = \begin{bmatrix} \Sigma & \mathsf{M}_{12} \\ \mathsf{M}_{21} & \mathsf{M}_{22} \end{bmatrix}$$

Extra Technical Background: 1D Example Cont.

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Obtain simulation

$$g(\mathcal{S}^G) = \mathcal{F}^{-1}(D^{1/2}\mathcal{F}(\mathbf{w}))\,[1:M]$$